

HTTP/HTTPS + TLS + Status Codes

6:30 PM to 7:30 PM -- 60 Minutes

SCHEDULE -- 6:30 to 7:30 PM

6:30-6:33	Skim all headings (3 min)
6:33-6:50	TOPIC 1 -- HTTP and HTTPS (17 min)
6:50-7:08	TOPIC 2 -- TLS Handshake (18 min)
7:08-7:22	TOPIC 3 -- HTTP Status Codes (14 min)
7:22-7:27	Final Rapid Fire (5 min)
7:27-7:30	Final Recall (3 min)

TOPIC 1 -- HTTP and HTTPS

What is HTTP?

HTTP stands for HyperText Transfer Protocol. It is the language your browser and a website's server use to talk to each other.

It is a request-response protocol. Browser sends a REQUEST. Server sends back a RESPONSE.

HTTP is stateless -- the server doesn't remember you between two separate requests. Each request is treated as brand new.

HTTP runs on top of TCP. It uses port 80 by default.

What is HTTPS and Why Do We Need It?

HTTPS = HTTP + Encryption (using a protocol called TLS).

Plain HTTP sends data in readable text. If someone intercepts it on the network, they can read your password, card number, everything.

HTTPS encrypts the data before sending. Even if intercepted, it looks like random garbage to the attacker.

HTTPS uses port 443 by default. You see a padlock icon in your browser when a site uses HTTPS.

The Exact Pipeline -- What Happens When You Visit a Banking Website

STEP 1: You type your bank's website URL in the browser.

STEP 2: Browser sees 'https' and knows it needs an encrypted connection.

STEP 3: Browser and server perform a TLS handshake to set up encryption (covered in Topic 2 below).

STEP 4: Once encryption is set up, browser sends an HTTP GET request -- but now it travels inside the encrypted tunnel.

STEP 5: Server processes the request and sends back the webpage as an HTTP response -- also encrypted.

STEP 6: Browser decrypts the response and shows you the page.

Without HTTPS at Step 3, anyone on the same WiFi network could read your bank login details in Step 4.

Feature	HTTP	HTTPS
Full form	HyperText Transfer Protocol	HTTP + TLS encryption
Security	Data sent as plain readable text	Data is encrypted
Port	80	443
Speed	Slightly faster (no encryption overhead)	Slightly slower (encryption adds small overhead)
Use today	Avoided for anything sensitive	Standard for all modern websites
Browser sign	No padlock, sometimes 'Not Secure' warning	Padlock icon shown

WHAT TO SAY IN INTERVIEW

HTTP is a stateless request-response protocol used by browsers and servers to communicate, but it sends data as plain readable text. HTTPS is HTTP combined with TLS encryption -- it encrypts the data so that even if intercepted, it cannot be read. HTTPS uses port 443 while HTTP uses port 80. Today, HTTPS is the standard for all websites, especially ones handling passwords or payments.

EXAMPLE: When you log into your college's exam portal using HTTP, your password travels as plain text -- anyone on the same WiFi network with the right tool could see it. If the portal uses HTTPS instead, your password is encrypted and unreadable even if intercepted.

- * HTTPS is a completely different protocol from HTTP -- WRONG. HTTPS is HTTP plus a TLS encryption layer.
- * HTTP is stateful -- WRONG. HTTP is stateless. Each request is independent. Cookies/sessions are used to fake statefulness.
- * HTTPS uses port 80 -- WRONG. HTTPS uses port 443. HTTP uses port 80.
- * HTTPS makes data unreadable forever -- it's encrypted in transit, but once decrypted at destination it's readable again.

RAPID FIRE

Q: HTTP port? -> 80

Q: HTTPS port? -> 443

Q: HTTPS = HTTP + ? -> TLS encryption

Q: Is HTTP stateless? -> Yes

Q: What shows padlock icon? -> HTTPS

Next Topic: TLS Handshake

TOPIC 2 -- TLS Handshake

What is TLS?

TLS stands for Transport Layer Security. It is the technology that makes HTTPS secure.

Before any encrypted data is sent, the browser and server must agree on HOW to encrypt. This agreement process is called the TLS Handshake.

Once the handshake is done, both sides share a secret key that only they know. All further data is encrypted using this shared key.

The Exact Pipeline -- TLS Handshake Step by Step

STEP 1 (Client Hello): Browser says 'I want a secure connection. Here are the encryption methods I support.'

STEP 2 (Server Hello): Server replies 'OK, let's use this encryption method. Here is my SSL Certificate (proves I am who I say I am).'

STEP 3 (Certificate Check): Browser checks if the certificate is valid and issued by a trusted authority. If fake or expired, browser shows a warning.

STEP 4 (Key Exchange): Browser and server now generate a shared SECRET KEY using the certificate's public key cryptography. Only the two of them know this key.

STEP 5 (Finished): Both sides confirm they have the same secret key. Handshake complete.

STEP 6 (Encrypted Data Transfer): From now on, ALL data between browser and server is encrypted using that secret key.

Step	What Happens	In Simple Words
1	Client Hello	Browser says hello and lists supported encryption methods
2	Server Hello + Certificate	Server replies and proves its identity with a certificate
3	Certificate Verification	Browser checks if the certificate is genuine
4	Key Exchange	Both sides create a shared secret key
5	Finished	Both confirm handshake is complete
6	Encrypted Communication	All data now travels encrypted

WHAT TO SAY IN INTERVIEW

TLS Handshake is the process where the browser and server agree on encryption before sending any actual data. It starts with the browser sending a Client Hello listing supported encryption methods, the server responding with its certificate, the browser verifying that certificate, both sides exchanging keys to create a shared secret, and finally confirming the handshake is complete. After this, all communication is encrypted using that shared secret key.

EXAMPLE: When you visit your bank's website, before your password is even typed, your browser and the bank's server already completed a TLS handshake -- exchanging a secret key. That's why the padlock icon appears even before you log in.

- * TLS Handshake happens after login -- WRONG. It happens before ANY data, even before the page loads.
- * TLS and SSL are completely different -- not quite. TLS is the modern, more secure version of SSL. People often use the terms interchangeably.
- * Certificate proves the connection is fast -- WRONG. Certificate proves the SERVER'S IDENTITY, not speed.
- * Same key is used by everyone -- WRONG. A unique secret key is generated for EACH connection.

RAPID FIRE

- Q: TLS full form? -> Transport Layer Security
- Q: What proves server identity? -> SSL Certificate
- Q: First message in handshake? -> Client Hello
- Q: Result of handshake? -> Shared secret key
- Q: After handshake, data is? -> Encrypted

Next Topic: HTTP Status Codes

TOPIC 3 -- HTTP Status Codes

What is a Status Code?

Every time a server responds to your browser's request, it includes a 3-digit status code.

This code tells you what happened -- success, error, redirect, or something else.

They are grouped by their first digit. You don't need to memorize every code -- just know the groups and the most common specific codes.

The 5 Groups -- Know These First

1xx (Informational): Request received, still processing. Rarely discussed in interviews.

2xx (Success): Everything worked.

3xx (Redirection): You need to go somewhere else to get what you want.

4xx (Client Error): YOU (the client) made a mistake in the request.

5xx (Server Error): The SERVER messed up, not you.

Code	Name	What It Really Means
200	OK	Request succeeded. Here is your data.
201	Created	New resource was successfully created (used after POST)
301	Moved Permanently	This URL has permanently moved to a new one
302	Found (Temporary Redirect)	Temporarily go to a different URL
400	Bad Request	Your request was malformed -- you sent something wrong
401	Unauthorized	You are not logged in. Please authenticate first
403	Forbidden	You ARE logged in, but you don't have permission for this
404	Not Found	The resource you asked for doesn't exist
500	Internal Server Error	Something broke on the server's side
503	Service Unavailable	Server is overloaded or down for maintenance

401 vs 403 -- The Most Confused Pair

401 Unauthorized: You are NOT logged in. The server doesn't know who you are. 'Please log in first.'

403 Forbidden: You ARE logged in (server knows who you are), but you don't have permission. 'I know you, but you can't access this.'

Real example: Trying to view someone else's private profile while not logged in = 401. Trying to view it while logged in as a regular user (not admin) = 403.

WHAT TO SAY IN INTERVIEW

HTTP status codes are 3-digit codes the server sends with every response to indicate the outcome. 2xx means success, 3xx means redirection, 4xx means the client made an error, and 5xx means the server had an error. The most commonly confused pair is 401 and 403 -- 401 means you're not authenticated at all, while 403 means you are authenticated but don't have permission to access that resource.

EXAMPLE: On an e-commerce site, if you try to access your order history without logging in, you get 401. If you're logged in as a normal customer but try to access the admin dashboard, you get 403 -- the server knows who you are, but denies permission.

- * 401 and 403 are the same -- WRONG. 401=not logged in. 403=logged in but no permission.
- * 404 means server crashed -- WRONG. 404 means the resource doesn't exist. 500 means server crashed.
- * All 4xx and 5xx mean the same severity -- WRONG. 4xx=client's fault. 5xx=server's fault. Different root cause.
- * 201 and 200 are interchangeable -- not quite. 200=general success. 201=specifically used after creating something with POST.

RAPID FIRE

- Q: 200 means? -> Success
- Q: 404 means? -> Not Found
- Q: 401 means? -> Not authenticated -- not logged in
- Q: 403 means? -> Authenticated but no permission
- Q: 500 means? -> Server error
- Q: 201 used after? -> Successful creation -- POST request

FINAL RAPID FIRE -- All Topics

- Q: HTTP port? -> A: 80
- Q: HTTPS port? -> A: 443
- Q: HTTPS = HTTP +? -> A: TLS encryption
- Q: TLS full form? -> A: Transport Layer Security
- Q: First TLS handshake message? -> A: Client Hello
- Q: What proves server identity? -> A: SSL Certificate
- Q: Result of TLS handshake? -> A: Shared secret key
- Q: 2xx means? -> A: Success
- Q: 4xx means? -> A: Client error
- Q: 5xx means? -> A: Server error

Q: 401 vs 403? -> A: 401=not logged in, 403=no permission

Q: 404 means? -> A: Resource not found

FINAL RECALL -- Close PDF. Write from memory: 1. HTTP vs HTTPS -- 3 differences 2. TLS Handshake -- all 6 steps in order 3. 5 groups of status codes -- one line each 4. 401 vs 403 -- the exact difference with example All correct = Done. Any wrong = re-read only that concept.

CN Day 5 -- HTTP/HTTPS + TLS + Status Codes Tomorrow: Day 6 -- How the Internet Works (google.com) + Routing + ARP